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TME–408

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

APPLIED THERMODYNAMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

(iii) Draw P-v and T-s diagrams whenever applicable.

1. (a) Derive the relation for efficiency of Diesel cycle in terms of compression ratio and cut-off ratio. (CO1)

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OR

(b) An air standard duel cycle has a compression ratio of 16 and compression begins at 1 bar, 50°C. The maximum pressure is 70 bar. The heat transfer to air at constant pressure is equal to that at constant volume. Estimate (i) the pressure and temperature at the cardinal points of the cycle, (ii) the cycle efficiency, (iii) the m.e.p. of cycle. Given $C_v = 0.718$ kJ/kg-K, $C_p = 1.005$ kJ/kg-K. (CO1)

2. (a) What are the *four* processes which constitute the Stirling cycle ? Show that the regenerative Stirling cycle has the same efficiency as the Carnot cycle. (CO1)

OR

(b) Explain the construction and working of Brayton cycle with regeneration, intercooling and reheating. Also represent the cycle on P-v and T-s diagram. (CO1)

3. (a) Explain the performance characteristics of an IC engine. What are the advantages and application of IC engines ? (CO1,CO2)

OR

(b) In a gas turbine plant working on the Brayton cycle with a regenerator of 75% effectiveness, the air at the inlet to the compressor is at 0.1 MPa, 30°C, the pressure ratio is 6 and maximum cycle temperature is 900°C. If the turbine and compressor have each an efficiency of 80%, find the percentage increase in cycle efficiency due to regeneration.(CO1, CO2)

4. (a) Derive the expression of optimum pressure ratio for a gas turbine plant having compressor efficiency η_c and turbine efficiency η_T. (CO2)

OR

(b) Define ramjet, scramjet, rocket, afterburner and turbofan. (CO2)

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5. (a) A gas turbine plant operates on the Brayton cycle between T_{min} = 300 K and T_{max} = 1037 K. Find the maximum work done per kg of air and the corresponding cycle efficiency. How does this efficiency compare with the Carnot cycle efficiency operating between the same two temperature ? (CO2)

OR

(b) Describe with neat sketches the working of a simple constant pressure open cycle gas turbine. (CO2)

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